2024 Introduction to VLSI Design
Homework 3: 4-bit Multiplexer-based Adder

Reconstructed and electrically verified solution

Simulation: 13 October 2026

1 Task and final design

Implement Figure 1 of the assignment using CIC018 TT, $V_{DD} = 1.8$ V, $W \geq 0.47\,\mu$m, and $L \geq 0.18\,\mu$m. Preserve the supplied pin order, input PWL waveforms, $C_{IN} = 0$, eight ordered vectors, and 9 ns power window. Report the worst delay for those vectors, average power, energy, and $\sum W_i L_i$. No layout or extracted parasitics enter this HW3 result.

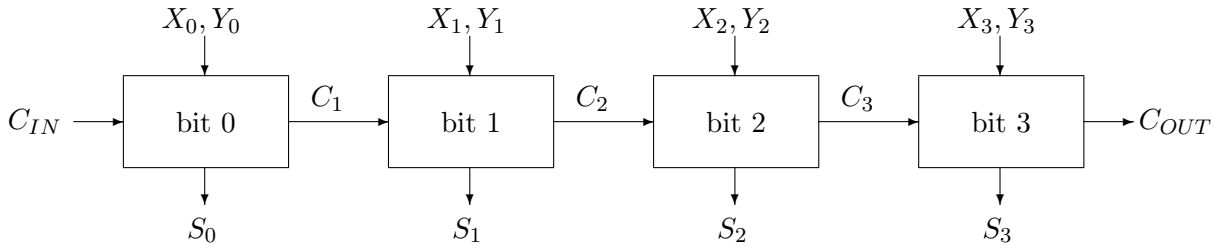
The selected design contains **64 MOSFETs**, all $W = 0.47\,\mu$m, $L = 0.18\,\mu$m. Define

$$e_i = \overline{X_i \oplus Y_i}, \tag{1}$$

$$S_i = \overline{e_i \oplus C_i} = X_i \oplus Y_i \oplus C_i, \tag{2}$$

$$C_{i+1} = e_i Y_i + \overline{e_i} C_i. \tag{3}$$

When $X_i = Y_i$, carry must equal Y_i; otherwise carry equals the incoming C_i. Thus the mux equation implements the full-adder carry exactly. All upper XNORs evaluate in parallel, while carry traverses four serial muxes. This follows the assignment's serial-parallel diagram; it has no clocked bit-serial operation or lookahead carry network.



Cell structure and area

Each bit uses a 4T upper XNOR, a 2T inverter producing $\overline{e}_i$, a 2T inverter producing $\overline{C}_i$, two 2T transmission gates for sum, and two 2T transmission gates for carry: $4+2+2+4+4 = 16$ transistors.

$$A_{\text{gate}} = 4(16)(0.47)(0.18) = \boxed{5.4144\,\mu\text{m}^2}.$$

This is the specified gate-area proxy, not a placed-and-routed footprint. For comparison, the supplied HW3 88T, $L = 0.24\,\mu$m topology has $A = 88(0.47)(0.24) = 9.9264\,\mu\text{m}^2$.

2 Transistor implementation

The compact upper XNOR can have a threshold-degraded *internal* high when $X = Y = 1$. The lower XNOR uses $C_i/\overline{C}_i$ as its *data* and $e_i/\overline{e}_i$ as its *select*; it does not pass the degraded e_i directly to S_i. The complementary CMOS inverter and PMOS/NMOS transmission-gate pair allow the external outputs to approach the rails. The actual TT electrical results, rather than an ideal Boolean assumption, establish the output levels below. This design has not been characterized for other corners, supplies, loads, or mismatch.

The following complete DUT source is included separately as `four_bit_multbased_adder.sp`. The testbench provides global rails and loads the licensed model library; the model is not part of the repository.

```
* Figure 1 serial-parallel multiplexer-based adder, 64 MOSFETs.
* Dimensions in micrometres; global rails vdd/vss provided by testbench.
.param WN=0.47u WP=0.47u WT=0.47u LG=0.18u
.subckt inv a z
MP z a vdd vdd P_18 W=WP L=LG
MN z a vss vss N_18 W=WN L=LG
.ends inv
.subckt tg a z s sb
MP z sb a vdd P_18 W=WT L=LG
MN z s a vss N_18 W=WT L=LG
.ends tg
.subckt mux_fa x y ci s co
* Compact 4T upper XNOR: e can be threshold-degraded internally.
MPU0 ep x vdd vdd P_18 W=WP L=LG
MPU1 e y ep vdd P_18 W=WP L=LG
MNU0 e y x vss N_18 W=WN L=LG
MNU1 e x y vss N_18 W=WN L=LG
XI2 e en inv
XI3 ci cib inv
* Lower XNOR uses ci/cib as DATA; e/en as SELECT.
* This choice restores the external sum even when compact e is degraded.
XT2 ci s e en tg
XT3 cib s en e tg
* Carry: co = e ? y : ci. Shared complementary select.
XT4 y co e en tg
XT5 ci co en e tg
.ends mux_fa
.subckt four_bit_multbased_adder cin x0 x1 x2 x3 y0 y1 y2 y3 s0 s1 s2 s3 cout
XF0 x0 y0 cin s0 c1 mux_fa
XF1 x1 y1 c1 s1 c2 mux_fa
XF2 x2 y2 c2 s2 c3 mux_fa
XF3 x3 y3 c3 s3 cout mux_fa
.ends four_bit_multbased_adder
```

3 Measurement definitions and required-vector results

All results use PrimeSim HSPICE S-2021.09 win64, CIC018 TT, 25°C, and 1.8 V, with no added output capacitors. Power includes the complete 0–9 ns interval, including the initial zero-vector interval and glitches.

$$P_{\text{total}} = \text{AVG}(\texttt{POWER}), \quad P_{VDD} = \text{AVG}[-V(vdd)I(vvdd)], \quad E_9 = P_{\text{total}}(9\,\text{ns}).$$

HSPICE's total circuit POWER is the negative sum of the powers of all independent sources. Input PWL drivers can supply energy through TG data paths, so P_{total} differs from P_{VDD}. The assignment's `.meas tran pwr avg power` uses total power; substituting the smaller VDD-only value would change the grading metric.

Measurement setting	P_{total} (μW)	P_{VDD} (μW)	E_9 (fJ)
Original `.tran 0.1ns 9n`	69.002216	37.998479	621.019943
Refined: 1 ps print / maximum step	69.558498	37.432840	626.026482

The original deck leaves DELMAX automatic. Refined measurements use `.tran 1p 9n` and `.option delmax=1p`; the 0.806% total-energy difference is reported explicitly. Candidate comparisons and delays use the same refined setting. No PWL input time or voltage is changed.

Vector	X	Y	Expected $\{C_{OUT}, S_3, S_2, S_1, S_0\}$	Measured	Result
1	1	5	00110	00110	pass
2	15	13	11100	11100	pass
3	11	9	10100	10100	pass
4	4	8	01100	01100	pass
5	3	7	01010	01010	pass
6	12	14	11010	11010	pass
7	2	15	10001	10001	pass
8	6	10	10000	10000	pass

At 0.8 ns after each vector endpoint, all eight vectors pass $V_L \leq 0.2$ V and $V_H \geq 1.6$ V. The maximum sampled low is 0.00396849 V and minimum high is 1.7910275 V. These are actual finite-time voltage readings, not an assertion of mathematically exact 0/1.8 V outputs.

Causal delay and glitches

For vector k, all changed input ramps run from $k - 0.1$ to k ns; their 50% crossings occur at $k - 0.05$ ns. For every output whose expected bit changes, use its *last* 0.9 V crossing in that vector interval:

$$t_{pd} = t_{\text{out,last }50\%} - (k - 0.05\,\text{ns}).$$

The maximum is **152.910320 ps**, from $(2, 15) \rightarrow (6, 10)$, S_0 falling: $t_{in} = 7.950$ ns and $t_{out} = 8.102910320$ ns. This is the worst observed delay for the required sequence, not a global maximum over all possible input transitions. The output data contain 12 extra 50% crossings from glitches. The conservative 1 ps sampled bound for all outputs to settle in the correct 0.2/1.6 V bands is 394 ps relative to the common input 50% event; it is a different metric from t_{pd}.

4 Delay evidence and sizing comparison

Vector	Output	Edge	Input 50% (ns)	Last output 50% (ns)	Delay (ps)
1	S1	rise	0.950	0.985495	35.495
1	S2	rise	0.950	1.041877	91.877
2	S1	fall	1.950	2.053628	103.628
2	S3	rise	1.950	2.088414	138.414
2	COUT	rise	1.950	1.956704	6.704
3	S3	fall	2.950	3.020838	70.838
4	S3	rise	3.950	4.044057	94.057
4	COUT	fall	3.950	4.067813	117.813
5	S1	rise	4.950	4.981098	31.098
5	S2	fall	4.950	5.030657	80.657
6	COUT	rise	5.950	5.956614	6.614
7	S0	rise	6.950	7.040801	90.801
7	S1	fall	6.950	7.079615	129.615
7	S3	fall	6.950	7.057709	107.709
8	S0	fall	7.950	8.102910	152.910

The original 88T design uses independent select inverters in its lower XNOR and carry mux. Sharing the complement reduces this to 80T. An 8T full-swing upper XNOR is retained in the 80T candidates. A second family uses the compact 4T upper XNOR from the supplied 2024 HW4 schematic idea, reducing the total to 64T. All six candidates pass the eight required vectors electrically under the same refined measurement conditions.

Candidate	$\sum WL$ (μm^2)	E_9 (fJ)	t_{pd} (ps)	Pt_{pd} (fJ)
88T reference-equivalent	9.9264	1005.327	247.239	27.617
80T minimum	6.7680	824.288	137.009	12.548
80T inverter P=0.94	8.1216	947.745	133.867	14.097
80T TG W=0.70	8.7552	1055.585	161.400	18.930
64T minimum (selected)	5.4144	626.026	152.910	10.636
64T P=0.70	6.0768	676.239	149.882	11.262

The selected minimum-size 64T circuit has the smallest area, energy, and power–delay product in this finite candidate set. Relative to the reference-equivalent 88T circuit, energy decreases by 37.73%, area by 45.45%, and measured delay by 38.15%. The minimum-size 80T circuit is faster than the 64T circuit, but consumes more energy and area and has a higher power–delay product. Increasing PMOS or TG sizes does not improve the final combined choice. This comparison does not establish a global optimum over all transistor networks or continuous sizes.

5 Electrical exhaustive verification and reproduction

In addition to the supplied sequence, the selected transistor circuit was simulated for all $16\times16\times2 = \mathbf{512}$ combinations of X, Y, and C_{IN}. Each vector occupies 2 ns, with 0.1 ns input ramps; voltages are sampled 1.8 ns after the endpoint. All **512/512** cases produce the arithmetic result $X + Y + C_{IN}$ and satisfy $V_L \leq 0.2$ V, $V_H \geq 1.6$ V on all five external outputs. Across those samples, $\max V_L = 0.002939216$ V and $\min V_H = 1.79601867$ V. This is transistor-level transient sampling of every settled input combination, not just Boolean software evaluation. It does not exhaust all 512^2 transitions or prove global worst delay.

`simulate.py` generates six candidate decks, runs licensed HSPICE, exports measured values, selects the 64T circuit, verifies the 512 electrical cases, and runs the original instructor testbench separately. Set `CIC018_MODEL` or `--model` to the licensed library and `HSPICE` or `--hspice` to the simulator executable.

```
python simulate.py --model D:/cic018.l --work .simulation --results results
```

The default Windows HSPICE installation is supplied in the script and can be overridden. All transistor netlists use the required `P_18`/`N_18` models and the exact assignment subcircuit pin order. The standalone original testbench changes only the model path from the instructor's Linux path to `D:/cic018.l`.

Output file	Evidence or purpose
`results/candidates.csv`	Dimensions, device count, area, total and supply power, energy, delay, objective products.
`results/vectors.csv`	Eight required vector results with five actual output voltages.
`results/delays.csv`	Causal input and output crossing times for every expected-changing output.
`results/settling.csv`	Per-vector output crossing counts and sampled band-settling bounds.
`results/exhaustive_512.csv`	All input combinations, arithmetic result, actual voltages, pass flags.
`results/waveform.csv`	1 ps sampled output voltages and signed VDD current; GUI screenshots are unnecessary.
`results/summary.json`	Conditions, model SHA256 identity, limits and result summary.
`results/instructor_exact.json`	Original 100 ps print-step / automatic DELMAX power result.

Sources reviewed and integration decisions

1. Jin-Fu Li, EE NCU, *EE3032 Introduction to VLSI Design, Homework 3*, Fall 2024; supplied Figure 1, original testbench and explanatory notes determine the topology, stimuli, minimum width, and grading window.

2. User-supplied 2024 HW3 reference by Chih-Hua Chang, archive containing `four_bit_multbased_adder.sp` and report. Its 88T TG design is reconstructed for equal-condition baseline comparison; misleading XOR comments in that source do not change its actual XNOR wiring.

3. User-supplied 2024 HW4 reference by Chih-Hua Chang, schematic `four_bit_multbased_adder.sp`. Its compact upper-XNOR idea is integrated after new electrical validation; neither its old measurements nor extracted layout parasitics are reused as HW3 results.

4. Public source: dOwOb1212, NCU_VLSI_Design_2023_HW, hw1/CLA_cmos.sp. Despite its CLA label, carry is passed serially through muxes. The source is 8-bit, uses widths below the 2024 minimum, and has different stimuli; it is an architectural cross-check, not a valid drop-in 2024 submission.

5. Local Synopsys S-2021.09 documentation: *Selecting Simulation Output Parameters*, `.OPTION DELMAX`, and `.OPTION LIS_NEW`. These define total POWER, internal time-step control, and machine-readable `.printtr0` output.

All numerical results above were newly simulated on 13 October 2026. The public repository contains derived statistics and source only; the licensed process library, student original reports, and raw license logs remain local.